Procedure – Generative AI Test Case Generation

Step 1

Open a Generative AI tool.

Step 2

Enter the requirement.

Step 3

Use the following prompt:
Analyze this software requirement:

A registered user should be able to log in using a valid username and password. Invalid credentials should be rejected and an appropriate error message should be displayed.

Generate:

1. **Positive test cases**
2. **Negative test cases**
3. **Test data**
4. **Expected results**
5. **Regression test scenarios**
6. **Possible defect-prone areas**
7. **Selenium Python automation code The final Selenium code should be executable in PyCharm.**

Step 4

Review the AI output.

Step 5

Check:

- Test cases
- Test data
- Expected results
- Selenium locators
- Python syntax
- Assertions
- Error handling

Step 6
Copy the reviewed and corrected Selenium code into PyCharm.

Step 7
Execute the Python file.

Source Code

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()
try:
driver.get("https://www.saucedemo.com/")
wait = WebDriverWait(driver, 10)
username = wait.until(
 EC.visibility_of_element_located(
 (By.ID, "user-name")

)
 )
password = driver.find_element(
 By.ID, "password"
)

login_button = driver.find_element(
By.ID, "login-button"
 )

username.send_keys("standard_user")
 password.send_keys("secret_sauce")
login_button.click()
```

```
wait.until(
EC.url_contains("inventory")
)

assert "inventory" in driver.current_url

print("AI-Assisted Selenium Test: PASS")
 print("Requirement Validation: PASS")
except AssertionError:
 print("AI-Assisted Selenium Test: FAIL")

finally:
 driver.quit()
```

Screenshot:

